

5. (a) (i) Let T be the tension.
 $2T \cos 75^\circ = 60$ 1M
 $T = 115.9 \text{ N}$ 1A 2
- (ii) energy stored in the string = k.e. of arrow 1M
 $= \frac{1}{2}(0.2)(45)^2$ 1A 2
 $= 202.5 \text{ J}$
- (b) (i) $d = v \cos 20^\circ t$ 1M
 $60 = 45 \cos 20^\circ t$
 $t = 1.42 \text{ s}$ 1A 2
- (ii) $h = 25 - \frac{1}{2}gt^2$ 1M
 $= 25 - \frac{1}{2}(9.81)(1.42)^2$
 $= 15.1 \text{ m [or } h = 14.9 \text{ m]}$ 1A 2